Dynamics from Probability:
Predictive Operators and the Koopman Picture

Patrick S. Mahon

Abstract

An observer sees a current state Q and a future state F. The classical approach to dynamics, following Koopman, describes the evolution of observable functions under a known transformation T — but this presupposes access to T. The prior question is: what does the observer *predict* about F given Q, without assuming the dynamics?

Making prediction precise forces an answer: the *predictive kernel* Π_Q is the regular conditional distribution of F given Q. Its existence requires disintegrating a joint distribution — which requires a σ-additive probability measure on the observable algebra. That measure is not assumed here; it is the object constructed in Paper I from coherent structured observations satisfying collective exhaustion.

Once Π_Q exists, its canonical minimal form is the *minimal predictive state map* Q_*, sending each sample point to its conditional law. The *predictive factorization theorem* shows that Q_* is the minimal sufficient statistic for future prediction. Indexing over time, the prediction laws compose, and their composition rule — the Chapman–Kolmogorov equation — is derived from temporal coherence rather than assumed. From it emerge the *predictive operators* $\{K_t\}$, their semigroup law, and the Koopman–Perron duality. When predictions are certain, the construction recovers the classical Koopman operator exactly.

The output is the *observable dynamical system* $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$. Setting $Q_t = h \circ T^t$ identifies Q_* with the delay map Φ_h, making this the direct input to the reconstruction question of Paper III.

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1 Introduction

The classical approach to dynamics, following Koopman [1], describes how observable functions evolve: the operator $U_T f = f \circ T$ acts by precomposing with a known transformation T. This presupposes access to T as a primitive. The question this paper asks is prior: given a current observation $Q : \Omega \to \alpha$ and a future observation $F : \Omega \to \beta$, what does the observer *predict* about F given Q? The dynamics, if there are any, should follow from the answer — not precede it.

Making prediction precise immediately generates a requirement. The canonical answer is the *predictive kernel* Π_Q: the regular conditional distribution of F given Q, the unique Markov kernel realising $P(F \in \cdot \mid Q = q)$ simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem, which in turn requires a σ-additive probability measure on the observable σ-algebra. That measure is not an extra assumption here. It is what Paper I constructs: a family of coherent finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion, the condition ruling out persistent mass on globally vanishing events. The predictive programme cannot begin until that measure exists; Paper I is the condition under which it does.

Once Π_Q exists, its minimal form is the *minimal predictive state map* $Q_* : \Omega \to \mathcal{P}(\beta)$, sending each sample point to its conditional law. The predictive factorization theorem shows that every conditional expectation of a function of F factors through Q_*: it is the minimal sufficient statistic for future prediction. Indexing over time, the prediction laws compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. This yields the *predictive operators* $\{K_t\}$, their semigroup law, and the Koopman–Perron duality. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

These structures assemble into the *observable dynamical system* $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$, which Paper III takes as its starting point and asks: when does the delay orbit of a single observable reconstruct the full state space?

2 Setup

The paper is self-contained. The connection to Paper I [2] is motivational; the formalism starts here from a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \to \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \to \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [3] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β, equipped with its canonical (Giry) σ-algebra. The *joint pushforward* is the measure $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$; its marginals are $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The main theorems are formalized in Lean 4 / Mathlib [4] in `PredictiveState.lean` and `PredictiveOperators.lean`. Several further results — the experiment preorder, general predictive sufficiency, the continuous-time generator — are stated in the paper but not yet formalized; these are noted at the relevant points.

3 The predictive kernel and the minimal predictive state map

3.1 The predictive kernel

Definition 3.1 (Predictive kernel)**.** The *predictive kernel* $\Pi_Q : \alpha \to \mathcal{P}(\beta)$ is the regular conditional distribution of F given Q:

$$\Pi_Q(q, A) \;=\; P(F \in A \mid Q = q),$$

defined as the disintegration kernel $\Pi_Q(q, \cdot) = \rho(\cdot \mid \alpha)$ of the joint pushforward $\rho = P \circ (Q, F)^{-1}$ over its first marginal.

Concretely: Π_Q is the unique (up to $P \circ Q^{-1}$-null sets) Markov kernel $\alpha \to \mathcal{P}(\beta)$ such that

$$\int_A \Pi_Q(q, B)\, d(P \circ Q^{-1})(q) \;=\; P(Q \in A,\ F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Proposition 3.2 (Predictive compatibility)**.** *Let $\pi : \alpha \to \alpha'$ be measurable with $Q' = \pi \circ Q$. Then for $(P \circ Q^{-1})$-almost every $q \in \alpha$:*

$$\Pi_Q(q, \cdot) \;=\; \Pi_{Q'}(\pi(q), \cdot).$$

Proof. The joint pushforward $\rho' = P \circ (Q', F)^{-1}$ satisfies $\rho' = \rho \circ (\pi \times \mathrm{id})^{-1}$. The claim follows from the $P \circ Q^{-1}$-a.e. uniqueness of the disintegration of ρ' over its first marginal, combined with the covariance of condKernel under measurable maps (`eq_condKernel_of_measure_eq_compProd` in Mathlib). □

3.2 The predictive law map and the minimal predictive state map

The predictive kernel Π_Q maps each value $q \in \alpha$ to a probability measure on outcomes. Packaging this as a single measurable map gives:

Definition 3.3 (Predictive law map and minimal predictive state map)**.** The *predictive law map* is

$$\varphi_Q : \alpha \to \mathcal{P}(\beta), \qquad \varphi_Q(q) = \Pi_Q(q, \cdot).$$

The *minimal predictive state map* is the composition

$$Q_* := \varphi_Q \circ Q : \Omega \to \mathcal{P}(\beta).$$

The value $Q_*(\omega)$ is the full conditional law $P(F \in \cdot \mid Q = Q(\omega))$.

Proposition 3.4 (Measurability)**.** *The maps φ_Q and Q_* are measurable.*

Proof. φ_Q is the underlying map of the Markov kernel Π_Q, which is measurable by definition of a kernel (`Kernel.measurable` in Mathlib). Since $\mathcal{P}(\beta)$ carries the subspace σ-algebra from the Giry space of all measures on β, the wrap as a `ProbabilityMeasure` is measurable by `Measurable.subtype_mk`. Measurability of Q_* then follows from composition. □

4 Predictive factorization

The predictive factorization theorem is the central result of this paper. It says that the conditional expectation of any bounded measurable function of F, given Q, factors through the minimal predictive state map Q_*.

Theorem 4.1 (Predictive factorization)**.** *For every bounded measurable $g : \beta \to \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] \;=\; \Big(\omega \mapsto \int g \, dQ_*(\omega)\Big) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g \, dQ_*(\omega) = \int g \, d\Pi_Q(Q(\omega), \cdot)$. We verify the two defining properties of the conditional expectation.

Measurability. The map $q \mapsto \int g \, d\Pi_Q(q, \cdot)$ is strongly measurable (Bochner integrability of g against a Markov kernel; `StronglyMeasurable.integral_kernel` in Mathlib). Since $Q_* = \varphi_Q \circ Q$, the map $h =$ (predictive operator applied to g) $\circ\, Q_*$ is $\sigma(Q)$-measurable.

Set-integral identity. For every $\sigma(Q)$-measurable set $S = Q^{-1}(A)$:

$$\begin{aligned}
\int_S h \, dP &= \int_{Q^{-1}(A)} \int g \, d\Pi_Q(Q(\omega), \cdot) \, dP(\omega) \\
&= \int_A \int g \, d\Pi_Q(q, \cdot) \, d(P \circ Q^{-1})(q) \quad \text{(push-forward)} \\
&= \int_A \int_{A \times \beta} g(f) \, d\rho(q, f) \quad \text{(disintegration identity)} \\
&= \int_{Q^{-1}(A)} g(F(\omega)) \, dP(\omega) = \int_S g(F) \, dP.
\end{aligned}$$

The key step is the disintegration identity for $\rho = P \circ (Q, F)^{-1}$, which in Mathlib is `Measure.setIntegral_condK`

By the a.e. uniqueness of conditional expectation (`ae_eq_condExp_of_forall_setIntegral_eq`), $h = E[g(F) \mid \sigma(Q)]$ P-a.e. □

Corollary 4.2 (Predictive sufficiency)**.** *$E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P-a.e.*

Proof. Since $Q_* = \varphi_Q \circ Q$, we have $\sigma(Q_*) \subseteq \sigma(Q)$. The conditional expectation given $\sigma(Q)$ already factors through Q_* by Theorem 4.1, so it is $\sigma(Q_*)$-measurable. □

The canonical predictive operator K_{Q_*} is the map

$$(K_{Q_*}g)(q_*) \;=\; \int g\, dq_*, \qquad q_* \in \mathcal{P}(\beta).$$

Theorem 4.1 then reads: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P-a.e. This is the canonical operator associated with the predictive state.

5 The predictive semigroup

So far, the setup has been static: a single pair of measurable maps (Q, F) representing present and future. To introduce dynamics, we index over a discrete time horizon. For each $t \in \mathbb{N}$, let $F_t : \Omega \to \gamma$ be the observable at time t, and let Π_t be the predictive kernel of F_t given the current predictive state.

5.1 Predictive kernel families

Definition 5.1 (Predictive kernel family and semigroup kernel)**.** A *predictive kernel family* on a state space γ is a family $\{\Pi_t\}_{t\in\mathbb{N}}$ of Markov kernels $\Pi_t : \gamma \to \mathcal{P}(\gamma)$ with $\Pi_0 = \delta$ (identity: $\Pi_0(q, \cdot) = \delta_q$).

A *predictive semigroup kernel* is a predictive kernel family satisfying the Chapman–Kolmogorov equation:

$$\Pi_{t+s}(q, \cdot) \;=\; \int_\gamma \Pi_t(q', \cdot)\, d\Pi_s(q, q') \quad \text{for all } q \in \gamma,\ t, s \in \mathbb{N}.$$

In kernel notation: $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ where $\circ_\kappa$ denotes kernel composition.

5.2 The predictive operator family

Definition 5.2 (Predictive operator family)**.** Given a predictive kernel family $\{\Pi_t\}$, the *horizon-indexed predictive operator* at horizon t is

$$(K_t g)(q) \;:=\; \int g\, d\Pi_t(q, \cdot), \qquad g : \gamma \to \mathbb{R}.$$

Proposition 5.3 (Basic properties)**.** *For any predictive kernel family $\{\Pi_t\}$:*

(i) Identity at zero*: $K_0 g = g$ for all measurable g;*

(ii) Positive*: $g \geq 0 \Rightarrow K_t g \geq 0$;*

(iii) Unital*: $K_t\mathbf{1} = \mathbf{1}$;*

(iv) Contractive*: $\|K_t g\|_\infty \leq \|g\|_\infty$.*

Proof. Identity at zero: $K_0 g(q) = \int g\, d\delta_q = g(q)$. Positivity and unitality are immediate from the Markov property and non-negativity of integration against a positive measure. Contractiveness: $|(K_t g)(q)| = |\int g\, d\Pi_t(q, \cdot)| \leq \|g\|_\infty \cdot \Pi_t(q, \gamma) = \|g\|_\infty$. □

Theorem 5.4 (Semigroup property)**.** *Let $\{\Pi_t\}$ be a predictive semigroup kernel. For every bounded measurable $g : \gamma \to \mathbb{R}$:*

$$K_{t+s} g \;=\; K_t(K_s g).$$

Proof. Using the Chapman–Kolmogorov equation $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$:

$$\begin{aligned}
(K_{t+s}g)(q) &= \int g \, d\Pi_{t+s}(q,\cdot) \\
&= \int g \, d(\Pi_t \circ_\kappa \Pi_s)(q,\cdot) \\
&= \int \Big(\int g \, d\Pi_t(q',\cdot) \Big) \, d\Pi_s(q,q') \quad \text{(Bochner–Fubini for kernel composition)} \\
&= \int (K_t g)(q') \, d\Pi_s(q,q') \;=\; (K_t(K_s g))(q).
\end{aligned}$$

The Bochner–Fubini step is `Kernel.integral_comp` from `Mathlib.Probability.Kernel.Composition.Integ`
The boundedness hypothesis on g is needed for integrability of g against the composed kernel. ☐

6 Koopman–Perron duality

The predictive operators K_t act on functions. The dual objects act on measures.

Definition 6.1 (Koopman–Perron dual)**.** Given a predictive semigroup kernel $\{\Pi_t\}$ and a (finite) measure μ on γ, the *Koopman–Perron push-forward* is

$$P_t^* \mu \;:=\; \mu \star \Pi_t \;=\; \int_\gamma \Pi_t(q,\cdot) \, d\mu(q).$$

In Mathlib notation: μ`.bind (ker` t`)`.

Theorem 6.2 (Koopman–Perron duality)**.** *For every bounded measurable $g : \gamma \to \mathbb{R}$ and finite measure μ on γ:*

$$\int_\gamma (K_t g) \, d\mu \;=\; \int_\gamma g \, d(P_t^* \mu).$$

Proof.

$$\begin{aligned}
\int g \, d(P_t^*\mu) = \int g \, d(\mu \star \Pi_t) \;&=\; \int g \, d\Big(\int \Pi_t(q,\cdot) \, d\mu(q) \Big) \\
&= \int_\gamma \int g \, d\Pi_t(q,\cdot) \, d\mu(q) \quad \text{(Bochner–Fubini)} \\
&= \int_\gamma (K_t g)(q) \, d\mu(q).
\end{aligned}$$

The Bochner–Fubini step again uses `Kernel.integral_comp`. ☐

7 Deterministic specialization

When the dynamics is deterministic, the stochastic picture collapses to the classical Koopman picture.

Theorem 7.1 (Deterministic specialization)**.** *Suppose $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \to \gamma$. Then for every measurable g:*

$$K_t g \;=\; g \circ \varphi_t.$$

Proof. $(K_t g)(q) = \int g \, d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The integral against a Dirac measure is `integral_dirac'` in Mathlib. ☐

Corollary 7.2 (Flow law)**.** *If* $\Pi_t(q,\cdot) = \delta_{\varphi_t(q)}$ *for all* t*, then*

$$\varphi_{t+s} \;=\; \varphi_t \circ \varphi_s.$$

Proof. From $\Pi_{t+s} = \Pi_t \circ_\kappa \Pi_s$ and the Dirac hypothesis:

$$\begin{aligned}\delta_{\varphi_{t+s}(q)} &= \Pi_{t+s}(q,\cdot) \;=\; (\Pi_t \circ_\kappa \Pi_s)(q,\cdot)\\ &= \Pi_s(q,\cdot) \star \Pi_t \;=\; \delta_{\varphi_s(q)} \star \Pi_t \;=\; \Pi_t(\varphi_s(q),\cdot) \;=\; \delta_{\varphi_t(\varphi_s(q))}.\end{aligned}$$

Since Dirac measures at distinct points are distinct (under `SeparatesPoints`), we conclude $\varphi_{t+s}(q) = \varphi_t(\varphi_s(q))$. □

The full structure produced by the construction is named once:

Definition 7.3 (Observable dynamical system)**.** An *observable dynamical system* is a triple $(O_{Q_*}, \nu_{Q_*}, \{K_t\}_{t\in\mathbb{N}})$ where $O_{Q_*} = \mathcal{P}(\beta)$ is the predictive state space, $\nu_{Q_*} = P \circ Q_*^{-1}$ is the predictive state distribution, and $\{K_t\}$ is the predictive operator family satisfying the semigroup law and Markov properties above.

8 Relation to Paper III

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction directly. Under this choice, $Q_* = \Phi_h$, the delay map $\Phi_h(x) = (h(T^n x))_{n\in\mathbb{Z}}$, and the predictive state space O_{Q_*} is precisely the delay orbit space. The observable dynamical system is the input to Paper III.

Paper III asks: when does the delay orbit of a single observable h determine the full state space? The answer is the *reconstruction theorem*: the observable σ-algebra $\mathcal{O}_h = \sigma(\{h \circ T^n\})$ equals $\mathcal{B}(X)$ modulo μ if and only if the algebra generated by $\{h \circ T^n\}$ is dense in $L^2(X,\mu)$. The proof turns on the density bridge: the L^2-closure of a function algebra equals the L^2-space of the σ-algebra it generates.

The Koopman operators $\{U_T^t\}$ connect the two papers. Paper II produces $\{K_t\}$ on O_{Q_*}; when $Q_* = \Phi_h$ is injective modulo null sets, K_t and U_T^t act on the same information. The cyclic vector condition — $\overline{\mathrm{span}}\{U_T^n h\} = L^2(X,\mu)$ — is sufficient for reconstruction but strictly stronger: reconstruction requires only that the *algebra* generated by the delay orbit be L^2-dense, not the linear span. They coincide only when U_T has simple spectrum.

When reconstruction holds, O_{Q_*} is the full state space X up to measure-theoretic identification. The compact Stone space introduced in Paper I as a technical device for measure extension is recognised by Paper III, under the reconstruction condition, as X itself. The programme begins with a compact completion of the sample space and ends by identifying it as the state space.

References

[1] Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.

[2] Patrick S. Mahon. Probability from observation: Carathéodory and Stone routes to the observable extension theorem. Preprint, 2026.

[3] Vladimir Abramovich Rokhlin. On the fundamental ideas of measure theory. *American Mathematical Society Translations*, 10:1–54, 1952.

[4] The Mathlib Community. Mathlib4: the Lean 4 mathematical library. `https://github.com/leanprover-community/mathlib4`, 2024.